

SPL1 Project Proposal Form, 2026
Institute of Information Technology (IIT)
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Project Description:

Title: NeuralForge - A Feedforward Neural Network Engine in Java

Neural networks are computational models made of layers of interconnected nodes. Each node computes a weighted sum of its inputs and passes it through a non-linear activation function. Learning occurs by comparing the network's output against the correct answer and propagating the error backwards through the layers to adjust weights - a process called backpropagation. This project implements such a system entirely from scratch in Java, with no external machine learning dependencies.

Features:

1. Configurable network architecture (number of layers and neurons per layer via config file)
2. Forward propagation and backpropagation
3. Activation functions: Sigmoid, ReLU, Softmax (selectable per layer)
4. Loss functions: Mean Squared Error and Cross-Entropy
5. Mini-batch gradient descent with configurable learning rate and batch size
6. Model save/load - serialize trained weights to file for later inference
7. Training from MNIST dataset
8. Training progress display (loss and accuracy per epoch)
9. GUI: draw a digit on a canvas with the mouse and see the network classify it in real time with confidence scores for each digit

Algorithms and Theories Used:

1. **Matrix Operations:** The entire forward and backward pass is expressed in terms of matrix multiplication, transposition, and element-wise operations. A custom matrix library will be built as the foundation.
2. **Feedforward Propagation:** Output at each layer is computed as $A = f(W \cdot X + b)$, applied sequentially from input to output layer.
3. **Backpropagation:** Error gradients are computed from the output layer backwards using the chain rule, determining each weight's contribution to the error.
4. **Mini-Batch SGD:** Gradients are averaged over small random batches and weights are updated as $W = W - \alpha \cdot \nabla W$.

5. **Xavier Weight Initialization:** Weights are initialized scaled to the number of input connections, preventing vanishing or exploding gradients early in training.

Languages or Tools to be used: Java, Swing, Git, Java standard libraries

Supervisor's Name: Dr. Emon Kumar Dey

Signature of the supervisor: _____

Date: _____

Before the Midterm Presentation:

I confirm that the progress is satisfactory and I am forwarding it for midterm presentation.

Signature of the supervisor: _____

Date: _____

Midterm Presentation Feedback:**Before the Final Presentation:**

I confirm that the project is complete and I am forwarding it for final presentation.

Signature of the supervisor: _____

Date: _____